Smart Student Survival System — Project Structure & Architecture

1. Project Structure

- **SmartStudentSurvival/**
  - **src/**
    - **model/**
      - Assignment.java
      - Exam.java
      - StudyTask.java
      - Course.java
      - Priority.java
    - **service/**
      - CGPAService.java
      - CountdownService.java
    - **util/**
      - FileHandler.java
      - DateUtil.java
    - **controller/**
      - AssignmentController.java
      - ExamController.java
      - PlannerController.java
      - CGPAController.java
    - **gui/**
      - Main.java
      - DashboardFrame.java
      - AssignmentPanel.java
      - ExamPanel.java
      - PlannerPanel.java
      - CGPAPanel.java
  - **data/**
    - assignments.txt
    - exams.txt
    - planner.txt
    - courses.txt
  - **docs/**
    - Project_Report.pdf
  - README.md

2. Layered Architecture (MVC + Services)

GUI Layer (View)
DashboardFrame, AssignmentPanel, ExamPanel, PlannerPanel, CGPAPanel
- User interface
- Swing components
- No direct file access

Controller Layer
AssignmentController, ExamController, PlannerController, CGPAController
- Handles user actions
- Keeps ArrayLists in memory

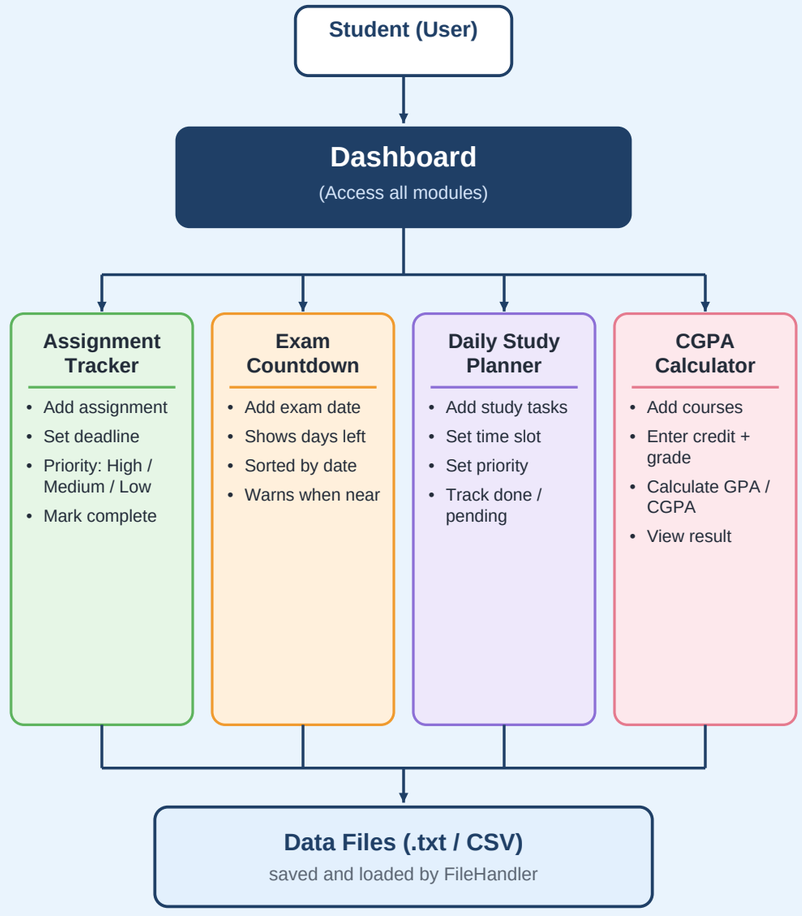
Service Layer
CGPAService, CountdownService
- Business logic
- CGPA calculation, days left
- Priority sorting

Model Layer
Assignment, Exam, StudyTask, Course, Priority
- Plain Java classes (objects)
- Fields, getters, setters

Util Layer
FileHandler, DateUtil
- File read / write
- Date parsing and checks

Data Layer
assignments.txt, exams.txt, planner.txt, courses.txt
- Text / CSV file storage
- Data survives restart

3. Modules Overview (System Flow)



4. Data File Format (Simplified)

assignments.txt
```
id
title
course
deadline
priority
status
```

exams.txt
```
id
subject
exam_date
room
syllabus
```

planner.txt
```
id
task
date
start_time
priority
done
```

courses.txt
```
id
course_name
credit
grade_point
semester
```

Each record is one line; fields are separated by commas (CSV).

- At startup FileHandler loads each file into an ArrayList; every add, edit or delete is saved back to the file.
- Priority values are High, Medium or Low (Priority enum). Grades convert to grade points for CGPA.

5. Key Features

- **Assignment Deadline Tracker** — Track every assignment with its due date
- **Exam Countdown** — Days left for each upcoming exam
- **Task Priority** — High, Medium, Low levels for tasks
- **Daily Study Planner** — Plan today's study tasks by time
- **Marks & CGPA Calculator** — Enter credits and grades, get CGPA
- **File Handling** — Auto save and load, no data lost

6. Technology Stack

Java	JDK 8 or higher
Swing	Desktop GUI
ArrayList	In-memory collections
java.io	File handling

Java concepts used

Classes, objects, methods, conditions, loops, ArrayList, file handling, Swing GUI

Future Expansion

Login screen, notes module, reminders, marks charts, search and filter

Smart Student Survival System | Plan. Study. Survive.